

Anomaly Detection & Time Series - Theory Answers

Q1. Anomaly Detection

Anomaly Detection is the process of identifying unusual patterns that differ from expected behavior. Types include Point Anomalies (single unusual observation), Contextual Anomalies (anomalous in a specific context), and Collective Anomalies (a group of observations that are anomalous together).

Q2. Isolation Forest vs DBSCAN vs LOF

Isolation Forest isolates anomalies using random partitions. DBSCAN identifies anomalies as points not belonging to dense clusters. LOF measures local density deviation. Isolation Forest is scalable, DBSCAN works well for arbitrary shapes, and LOF excels in local anomaly detection.

Q3. Components of Time Series

The main components are Trend (long-term movement), Seasonality (repeating patterns), Cyclical Variation (long-term oscillations), and Residuals (random noise).

Q4. Stationarity

A stationary time series has constant mean, variance, and autocorrelation over time. Tests include the Augmented Dickey-Fuller test. Transformations include differencing, log transformation, and detrending.

Q5. AR, MA, ARIMA, SARIMA, SARIMAX

AR uses past values, MA uses past errors, ARIMA combines both with differencing, SARIMA adds seasonal effects, and SARIMAX includes external variables for forecasting.

Q10. Real-Time Energy Monitoring Workflow

Use Isolation Forest or LOF for anomaly detection. Use SARIMAX for forecasting because it can incorporate weather and external variables. Validate using rolling forecasts, MAE, RMSE, and anomaly precision-recall. The solution improves grid stability, demand planning, outage prevention, and operational efficiency.